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Preparation of Multigrain Bread



Manmath Kornule (202401110045)
Krishna Kushwah (202401110039)
Om Channawar (202401110044)
Arjun Mankuskar (202401110038)

Introduction

- **Project Overview**: Preparation of Multigrain Bread, a nutritious bakery product combining grains like wheat, oats, millet, and barley.
- **Key Processes Demonstrated**:
  - Fermentation: Biological process using yeast.
  - Baking: Heat treatment step.

- **Main Objectives (Purpose of Prototype)**:
  - To understand the chemical and process engineering principles (mixing, fermentation, heat transfer) in food manufacturing.
  - To gain practical exposure to batch manufacturing, quality testing, and process optimization.



Raw Materials and Key Components

Total input mass: ≈ 455 g

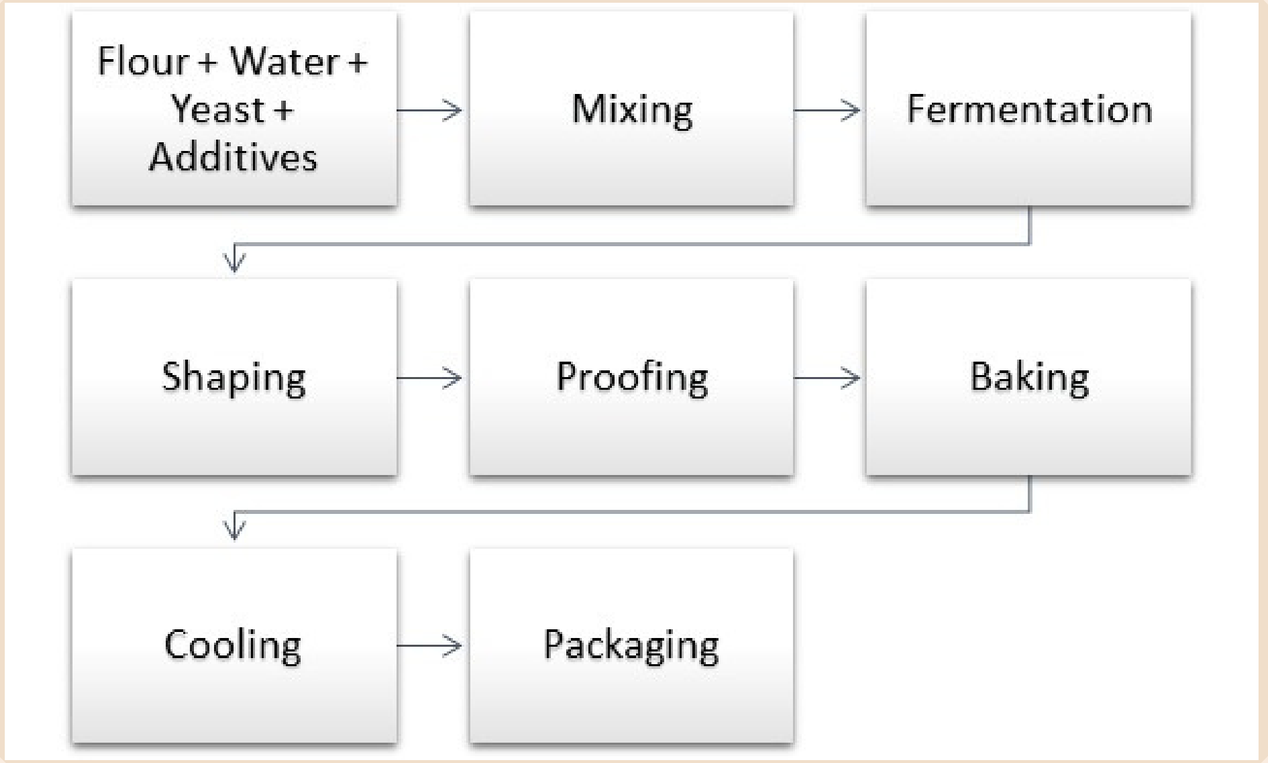
Material list (approx.):

- Multigrain flour (wheat, oats, millet): 250 g — base
- Water: 150 mL — hydration
- Yeast: 10 g — ferment
- Sugar: 20 g — feed for yeast
- Salt: 5 g — flavor
- Vegetable oil: 20 mL — softening

Source: Food-grade materials from local grocers.

Process Flow & Equipment





Experimental Procedure & Key Parameters

Steps and Key Settings:

• Weigh & Mix: Dry ingredients first, then add water and oil slowly.
• Knead: Until smooth and springy (10–12 minutes).
• Ferment: 1.5–2 hours at 30–35°C.
• Shape & Proof: 30–45 minutes.
• Bake: Preheated oven.

Best Results:

– Fermentation: 2 hours at 30–35°C
– Baking: 200°C for 30 minutes (better than the initial 180°C)
– Hydration: 60% (was 55%)

Results, and Quality Control

Methods Used:

- Visual inspection (color, crust, shape)
- Texture test (softness and elasticity)
- Taste & aroma (sensory evaluation)
- Moisture content (estimated by weight loss)

Results:

Bread was uniformly baked, with soft texture and pleasant aroma.
No cracks or dense areas were observed.

📊 Performance Comparison

Parameter	Expected	Obtained
Color	Golden brown	Golden brown
Texture	Soft and fluffy	Soft and uniform
Weight Loss	10–12%	10%

Challenges and Environmental Considerations

Challenges and Environmental Considerations

What We Faced & How We Solved It

• Uneven fermentation → Steady temperature and humidity.
• Dense texture from low hydration → Water ratio raised from 55% to 60%.
Key lesson: Fermentation needs careful timing and temperature.

Environmental & Safety

• Waste: Minimal dough residue, composted.
• Impact: Relies on local materials, safe for the environment, moderate energy use.

Cost Analysis

Item	Quantity	Cost (₹)
Multigrain flour	1Kg	71/-
Yeast	25 g	40/-
Sugar, Salt, Oil	-	40/-
Electricity / Fuel	—	10/-
Total Cost		₹161/-



- Economic Feasibility: The process is low-cost and feasible for small-scale or cottage food production units.
- Equipment Cost Note: Lab equipment (oven, balance) was institutional; depreciation/maintenance costs were not included in the batch total.



Conclusion

Summary

- The multigrain bread prototype was successfully developed, exhibiting good texture, appearance, and a high yield.

Improvements

- Test different grain combinations for better nutrition
- Scale up the batch to pilot scale for industrial testing
- Conduct shelf-life and microbial testing for extended storage

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Thank You Very Much